CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 9: Simple and Compound Interest

Topic-wise short notes with solved examples

Simple interest and its formula, finding the principal, rate or time, and compound interest calculated year by year and by formula, with the difference between the two.

1 Simple interest

- **Principal** P: the money borrowed or deposited. **Rate** R: the interest on ₹100 for one year, written as a percentage per annum. **Time** T: the period in years.
- **Interest** I: the extra money paid for using the principal. **Amount** $A = P + I$.

$$I = \frac{P \times R \times T}{100}$$

- Time must be in **years**: 6 months $= \frac{1}{2}$ year, 73 days $= \frac{73}{365} = \frac{1}{5}$ year.
- When counting days between two dates, leave out the first day and include the last.

Example 1.1 Find the simple interest and the amount on ₹6000 at 8% per annum for 3 years.

SOLUTION

$$I = \frac{6000 \times 8 \times 3}{100} = 1440$$

Interest ₹1440; amount ₹7440

Example 1.2 Find the simple interest on ₹7300 at 10% per annum for 146 days.

SOLUTION

$T = \frac{146}{365} = \frac{2}{5}$ year. $I = \frac{7300 \times 10 \times 2}{100 \times 5}$.

₹292

2 Finding P, R or T

To find	Formula
Principal	$P = \frac{100 \times I}{R \times T}$
Rate	$R = \frac{100 \times I}{P \times T}$
Time	$T = \frac{100 \times I}{P \times R}$
Principal from the amount	$P = \frac{100 \times A}{100 + R \times T}$

Example 2.1 At what rate will ₹4000 earn ₹1200 as simple interest in 5 years?

SOLUTION

$$R = \frac{100 \times 1200}{4000 \times 5}$$

6% per annum

Example 2.2 In how many years will ₹2500 amount to ₹3250 at 6% per annum simple interest?

SOLUTION

$I = 750.\ T = \frac{100 \times 750}{2500 \times 6}.$

5 years

Example 2.3 What sum amounts to ₹5800 in 4 years at 4% per annum simple interest?

SOLUTION

$$P = \frac{100 \times 5800}{100 + 16}$$

₹5000

Example 2.4 A sum doubles itself in 8 years at simple interest. Find the rate.

SOLUTION

Interest equals the principal: $P = \frac{P \times R \times 8}{100}.$

12.5% per annum

Example 2.5 A sum amounts to ₹5600 in 2 years and to ₹6400 in 3 years at simple interest. Find the sum and the rate.

SOLUTION

Interest for 1 year $= 800$. Interest for 2 years $= 1600$, so the sum is $5600 - 1600 = 4000$.

$$R = \frac{100 \times 800}{4000 \times 1}$$

₹4000; 20% per annum

3 Compound interest

- In **compound interest** the interest of each year is added to the principal, and the next year's interest is calculated on this larger sum.
- **Amount at the end of a year becomes the principal for the next year.**
- Compound interest (C.I.) = final amount − original principal.
- For the first year, C.I. and S.I. are equal. After that, C.I. is greater.

Year	Principal (₹)	Interest at 10% (₹)	Amount (₹)
1	8000	800	8800
2	8800	880	9680
3	9680	968	10648

Example 3.1 Find the compound interest on ₹8000 for 3 years at 10% per annum.

SOLUTION

From the table the amount is ₹10,648.

₹2648

Example 3.2 Find the amount on ₹5000 for 2 years when the rates for the two years are 8% and 10%.

SOLUTION

After year 1: $5000 + 400 = 5400$. After year 2: $5400 + 540$.

₹5940

4 Compound interest by formula

$$A = P\left(1 + \frac{R}{100}\right)^{n}, \qquad \text{C.I.} = A - P$$

- Here n is the number of years and the interest is compounded yearly.
- **Different rates**: $A = P\left(1 + \frac{R_1}{100}\right)\left(1 + \frac{R_2}{100}\right)$.
- **Compounded half-yearly**: use rate $\frac{R}{2}\%$ and $2n$ periods.

Example 4.1 Find the compound interest on ₹12,000 for 2 years at 5% per annum.

SOLUTION

$$A = 12000 \times \left(\frac{21}{20}\right)^{2} = 12000 \times \frac{441}{400} = 13230$$

₹1230

Example 4.2 Find the amount on ₹10,000 for 1 year at 10% per annum, compounded half-yearly.

SOLUTION

Two half-years at 5%: $10000 \times (1.05)^2$

₹11,025

Example 4.3 What sum amounts to ₹6050 in 2 years at 10% per annum compound interest?

SOLUTION

$P \times 1.21 = 6050$

₹5000

5 Comparing C.I. and S.I.

- For 2 years: C.I. $-$ S.I. $= P\left(\frac{R}{100}\right)^2$. It equals one year's interest on the first year's interest.

Example 5.1 Find the difference between the C.I. and the S.I. on ₹5000 for 2 years at 6% per annum.

SOLUTION

$5000 \times 0.06 \times 0.06$

₹18

Example 5.2 The difference between the C.I. and the S.I. on a sum for 2 years at 5% is ₹15. Find the sum.

SOLUTION

$P \times \frac{1}{400} = 15$

₹6000

6 Growth and depreciation

- A quantity growing by $R\%$ a year becomes $P\left(1 + \frac{R}{100}\right)^n$ after n years.
- A quantity falling by $R\%$ a year becomes $P\left(1 - \frac{R}{100}\right)^n$.

Example 6.1 A town of 20,000 people grows by 5% a year. Find its population after 2 years.

SOLUTION

20000×1.05^2

22,050

Example 6.2 A machine worth ₹50,000 loses 10% of its value each year. Find its value after 2 years.

SOLUTION

50000×0.9^2

₹40,500

7 Common mistakes

- Using months or days as T without converting to years.
- Giving the amount when the interest is asked, or the reverse.
- Calculating every year's compound interest on the original principal.
- Writing $(1+R)^n$ instead of $\left(1+\frac{R}{100}\right)^n$.
- For half-yearly compounding, halving the rate but not doubling the number of periods.
- Using a plus sign for depreciation.

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